# SAP Schema Documentation

SAP Business Network Freight Collaboration
Message Implementation Guide | Public
LBN_ANSI_X12_310
Message Type: 310
Document Version: 1.1 – 2026-05-16



Run Simple

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# 1 OVERVIEW HEADER

## General Information

**Name** ANSI X12 Interchange headers and trailers

**Direction** Out

**Status** Active

**Message Type** Interchange Structure

**External Category** Message

## Documentation

**Definition** The Interchange Envelope, often referred to as the “outer envelope,” is the wrapper for all the data to be sent in one transmission. It can contain multiple Functional Groups. This characteristic means that transactions of different types can be included in the Interchange Envelope, with each type of transaction stored in a separate Functional Group. The Interchange Envelope is defined by the header and trailer; the Interchange Control Header (designated ISA) appears at the beginning, and the Interchange Control Trailer (designated IEA) appears at the end. While the typical pattern from Enterprise Systems is to create one Functional Group (GS/GE) within an Interchange Group (ISA/IEA), the X12 enveloping supports one or more Functional Groups (GS/GE) within an Interchange Group (ISA/IEA).

## 1.1 Structure

The following table shows the complete structure.

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| ├ **S_ISA** — Interchange Control Header | 1 .. 1 | | | | |
| │ ├ **D_I01** — Authorization Information Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I01 |
| │ ├ **D_I02** — Authorization Information | 1 .. 1 | String | | 10 .. 10 | |
| │ ├ **D_I03** — Security Information Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I03 |
| │ ├ **D_I04** — Security Information | 1 .. 1 | String | | 10 .. 10 | |
| │ ├ **D_I05_1** — Interchange ID Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I05 |
| │ ├ **D_I06** — Interchange Sender ID | 1 .. 1 | String | | 15 .. 15 | |
| │ ├ **D_I05_2** — Interchange ID Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I05 |
| │ ├ **D_I07** — Interchange Receiver ID | 1 .. 1 | String | | 15 .. 15 | |
| │ ├ **D_I08** — Interchange Date | 1 .. 1 | String | | 6 .. 6 | |
| │ ├ **D_I09** — Interchange Time | 1 .. 1 | String | | 4 .. 4 | |
| │ ├ **D_I10** — Interchange Control Standards Identifier | 1 .. 1 | String | | 1 .. 1 | |
| │ ├ **D_I11** — Interchange Control Version Number | 1 .. 1 | String | | 5 .. 5 | D_I11 |
| │ ├ **D_I12** — Interchange Control Number | 1 .. 1 | String | | 9 .. 9 | |
| │ ├ **D_I13** — Acknowledgment Requested | 1 .. 1 | String | | 1 .. 1 | D_I13 |
| │ ├ **D_I14** — Interchange Usage Indicator | 1 .. 1 | String | | 1 .. 1 | D_I14 |
| │ └ **D_I15** — Component Element Separator | 1 .. 1 | String | | 1 .. 1 | |
| ├ **FunctionalGroup** — Functional Group | 1 .. 1 | | | | |
| │ ├ **S_GS** — Functional Group Header | 1 .. 1 | | | | |
| │ │ ├ **D_479** — Functional Identifier Code | 1 .. 1 | String | | 2 .. 2 | D_479 |
| │ │ ├ **D_142** — Application Sender's Code | 1 .. 1 | String | | 2 .. 15 | |
| │ │ ├ **D_124** — Application Receiver's Code | 1 .. 1 | String | | 2 .. 15 | |
| │ │ ├ **D_373** — Date | 1 .. 1 | String | | 8 .. 8 | |
| │ │ ├ **D_337** — Time | 1 .. 1 | String | | 6 .. 6 | |
| │ │ ├ **D_28** — Group Control Number | 1 .. 1 | String | | 9 .. 9 | |
| │ │ ├ **D_455** — Responsible Agency Code | 1 .. 1 | String | | 1 .. 1 | D_455 |
| │ │ └ **D_480** — Version / Release / Industry Identifier Code | 1 .. 1 | String | | 6 .. 6 | D_480 |
| │ └ **S_GE** — Functional Group Trailer | 1 .. 1 | | | | |
| │ ├ **D_97** — Number of Transaction Sets Included | 1 .. 1 | String | | 1 .. 1 | |
| │ └ **D_28** — Group Control Number | 1 .. 1 | String | | 9 .. 9 | |
| └ **S_IEA** — Interchange Control Trailer | 1 .. 1 | | | | |
| ├ **D_I16** — Number of Included Functional Groups | 1 .. 1 | String | | 1 .. 1 | |
| └ **D_I12** — Interchange Control Number | 1 .. 1 | String | | 9 .. 9 | |

# 2 DETAILS HEADER

## 2.1 Interchange — Interchange Structure

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| ├ **S_ISA** — Interchange Control Header | 1 .. 1 | | | | |
| ├ **FunctionalGroup** — Functional Group | 1 .. 1 | | | | |
| └ **S_IEA** — Interchange Control Trailer | 1 .. 1 | | | | |

### Documentation

**Definition** The Interchange Envelope, often referred to as the “outer envelope,” is the wrapper for all the data to be sent in one transmission. It can contain multiple Functional Groups. This characteristic means that transactions of different types can be included in the Interchange Envelope, with each type of transaction stored in a separate Functional Group. The Interchange Envelope is defined by the header and trailer; the Interchange Control Header (designated ISA) appears at the beginning, and the Interchange Control Trailer (designated IEA) appears at the end. While the typical pattern from Enterprise Systems is to create one Functional Group (GS/GE) within an Interchange Group (ISA/IEA), the X12 enveloping supports one or more Functional Groups (GS/GE) within an Interchange Group (ISA/IEA).

### Properties

**Identifier** Interchange

**Name** Interchange Structure

**Cardinality** min: 1 max: 1

## 2.2 S_ISA — Interchange Control Header

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| ├ **S_ISA** — Interchange Control Header | 1 .. 1 | | | | |
| │ ├ **D_I01** — Authorization Information Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I01 |
| │ ├ **D_I02** — Authorization Information | 1 .. 1 | String | | 10 .. 10 | |
| │ ├ **D_I03** — Security Information Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I03 |
| │ ├ **D_I04** — Security Information | 1 .. 1 | String | | 10 .. 10 | |
| │ ├ **D_I05_1** — Interchange ID Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I05 |
| │ ├ **D_I06** — Interchange Sender ID | 1 .. 1 | String | | 15 .. 15 | |
| │ ├ **D_I05_2** — Interchange ID Qualifier | 1 .. 1 | String | | 2 .. 2 | D_I05 |
| │ ├ **D_I07** — Interchange Receiver ID | 1 .. 1 | String | | 15 .. 15 | |
| │ ├ **D_I08** — Interchange Date | 1 .. 1 | String | | 6 .. 6 | |
| │ ├ **D_I09** — Interchange Time | 1 .. 1 | String | | 4 .. 4 | |
| │ ├ **D_I10** — Interchange Control Standards Identifier | 1 .. 1 | String | | 1 .. 1 | |
| │ ├ **D_I11** — Interchange Control Version Number | 1 .. 1 | String | | 5 .. 5 | D_I11 |
| │ ├ **D_I12** — Interchange Control Number | 1 .. 1 | String | | 9 .. 9 | |
| │ ├ **D_I13** — Acknowledgment Requested | 1 .. 1 | String | | 1 .. 1 | D_I13 |
| │ ├ **D_I14** — Interchange Usage Indicator | 1 .. 1 | String | | 1 .. 1 | D_I14 |
| │ └ **D_I15** — Component Element Separator | 1 .. 1 | String | | 1 .. 1 | |

### Documentation

**Definition** To start and identify an interchange of zero or more functional groups and interchange-related control segments

### Properties

**Identifier** S_ISA

**Name** Interchange Control Header

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Element

### Leaf Elements

| Identifier/Name | Description/Notes/Code Values | Properties |
|---|---|---|
| **D_I01**<br>Authorization Information Qualifier | Code identifying the type of information in the Authorization Information<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 2 max: 2<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I01<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I02**<br>Authorization Information | Information used for additional identification or authorization of the interchange sender or the data in the interchange; the type of information is set by the Authorization Information Qualifier (I01) | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 10 max: 10<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I03**<br>Security Information Qualifier | Code identifying the type of information in the Security Information<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 2 max: 2<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I03 |

| | | |
|---|---|---|
| | | **Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I04**<br>Security Information | This is used for identifying the security information about the interchange sender or the data in the interchange; the type of information is set by the Security Information Qualifier (I03) | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 10 **max:** 10<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I05_1**<br>Interchange ID Qualifier | Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 2 **max:** 2<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I05<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I06**<br>Interchange Sender ID | Identification code published by the sender for other parties to use as the receiver ID to route data to them; the sender always codes this value in the sender ID element<br><br>**Usage 1**<br>This field contains the sender LBN ID.<br>**Constraint 1**<br>For EDI 997 message type the LBN ID is always “LBN” as the message will be generated by the network. | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 15 **max:** 15<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I05_2**<br>Interchange ID Qualifier | Code indicating the system/method of code structure used to designate the sender or receiver ID element being qualified<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 2 **max:** 2<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I05<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I07**<br>Interchange Receiver ID | Identification code published by the receiver of the data; When sending, it is used by the sender as their sending ID, thus other parties sending to them will use this as a receiving ID to route data to them<br><br>**Usage 1**<br>This field contains the receiver LBN ID. | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 15 **max:** 15<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I08**<br>Interchange Date | Date of the interchange | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 6 **max:** 6<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I09**<br>Interchange Time | Time of the interchange | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 4 **max:** 4<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I10**<br>Interchange Control Standards Identifier | Code to identify the agency responsible for the control standard used by the message that is enclosed by the interchange header and trailer | **Cardinality** **min:** 1 **max:** 1<br>**Primitive Type** String<br>**Length** **min:** 1 **max:** 1<br>**External Category** Element<br>**Data Type** xsd:string |

| | | |
|---|---|---|
| **D_I11**<br>Interchange Control Version Number | Code specifying the version number of the interchange control segments<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 5 max: 5<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I11<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I12**<br>Interchange Control Number | A control number assigned by the interchange sender | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 9 max: 9<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_I13**<br>Acknowledgment Requested | Code indicating sender's request for an interchange acknowledgement<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 1<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I13<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I14**<br>Interchange Usage Indicator | Code indicating whether data enclosed by this interchange envelope is test, production or information<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 1<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_I14<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_I15**<br>Component Element Separator | Type is not applicable; the component element separator is a delimiter and not a data element; this field provides the delimiter used to separate component data elements within a composite data structure; this value must be different than the data element separator and the segment terminator | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 1 max: 1<br>**External Category** Element<br>**Data Type** xsd:string |

## 2.3 FunctionalGroup — Functional Group

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| ├ **FunctionalGroup** — Functional Group | 1 .. 1 | | | | |
| │ ├ **S_GS** — Functional Group Header | 1 .. 1 | | | | |
| │ └ **S_GE** — Functional Group Trailer | 1 .. 1 | | | | |

### Documentation

**Definition** Functional Groups, often referred to as the “inner envelope,” are made up of one or more Transaction Sets. One Functional Group Envelope must include transaction of all of the same type, which can be batched together into one transmission. The Functional Group is defined by the header and trailer segments.

### Properties

**Identifier** FunctionalGroup

**Name** Functional Group

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Element

## 2.4 S_GS — Functional Group Header

### Structure

| Node | Card. | Prim.Type | Pos. | Length | Codelists |
|---|---|---|---|---|---|
| **Interchange** — Interchange Structure | 1 .. 1 | | | | |
| ├ **FunctionalGroup** — Functional Group | 1 .. 1 | | | | |
| │ ├ **S_GS** — Functional Group Header | 1 .. 1 | | | | |
| │ │ ├ **D_479** — Functional Identifier Code | 1 .. 1 | String | | 2 .. 2 | D_479 |
| │ │ ├ **D_142** — Application Sender's Code | 1 .. 1 | String | | 2 .. 15 | |
| │ │ ├ **D_124** — Application Receiver's Code | 1 .. 1 | String | | 2 .. 15 | |
| │ │ ├ **D_373** — Date | 1 .. 1 | String | | 8 .. 8 | |
| │ │ ├ **D_337** — Time | 1 .. 1 | String | | 6 .. 6 | |
| │ │ ├ **D_28** — Group Control Number | 1 .. 1 | String | | 9 .. 9 | |
| │ │ ├ **D_455** — Responsible Agency Code | 1 .. 1 | String | | 1 .. 1 | D_455 |
| │ │ └ **D_480** — Version / Release / Industry Identifier Code | 1 .. 1 | String | | 6 .. 6 | D_480 |

### Documentation

**Definition** To indicate the beginning of a functional group and to provide control information Comments 1. A functional group of related transaction sets, within the scope of X12 standards, consists of a collection of similar transaction sets enclosed by a functional group header and a functional group trailer. Semantic Notes: 1. GS04 is the group date. 2. GS05 is the group time. 3. The data interchange control number GS06 in this header must be identical to the same data element in the associated functional group trailer, GE02.

### Properties

**Identifier** S_GS

**Name** Functional Group Header

**Cardinality** min: 1 max: 1

### Syntax Type Related

**External Category** Element

### Leaf Elements

| Identifier/Name | Description/Notes/Code Values | Properties |
|---|---|---|
| **D_479**<br>Functional Identifier Code | Code identifying a group of application related transaction sets<br><br>**Selected Values**<br>All Values in Codelist | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 2 max: 2<br>**External Category** Element<br>**Data Type** xsd:string<br><br>**Codelist Id** D_479<br>**Type System** Customer_TS<br>**Version Mode** Local<br>**Version** 1.1 |
| **D_142**<br>Application Sender's Code | Code identifying party sending transmission; codes agreed to by trading partners | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 2 max: 15<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_124**<br>Application Receiver's Code | Code identifying party receiving transmission; codes agreed to by trading partners | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 2 max: 15<br>**External Category** Element<br>**Data Type** xsd:string |
| **D_373**<br>Date | Date expressed as CCYYMMDD where CC represents the first two digits of the calendar year | **Cardinality** min: 1 max: 1<br>**Primitive Type** String<br>**Length** min: 8 max: 8<br>**External Category** Element |